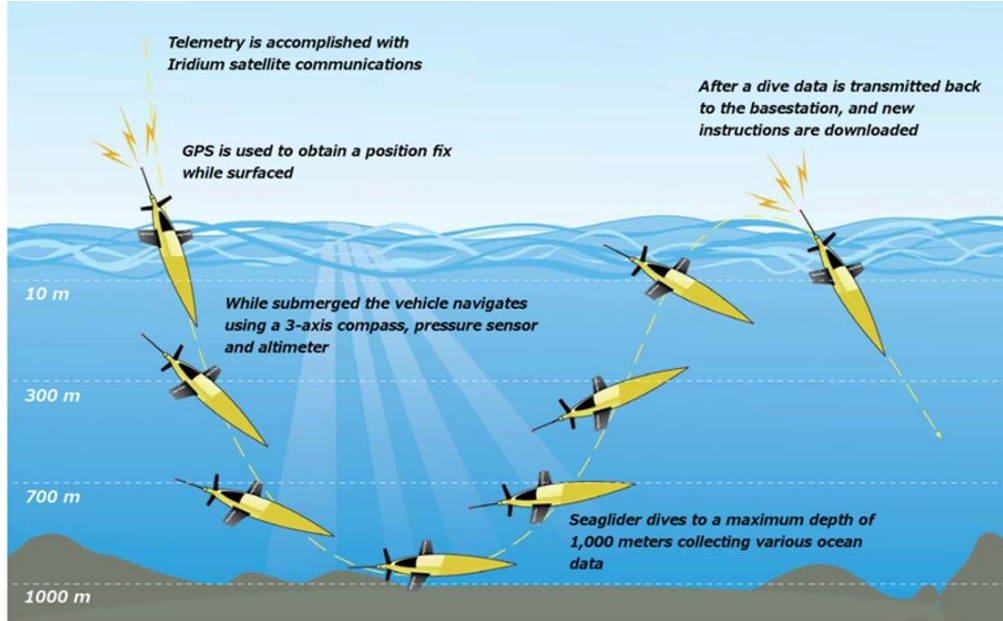


--- Glider operations and data



Ocean gliders 101



- **Buoyancy driven:** adjust volume of external bladder to sink or float
- **V shaped dives (“yo-yos”):** wings convert vertical force to horizontal lift
- **Pitch, roll, heading:** controlled by internal mass shifters or rudder
- **Low energy consumption = long endurance**

Operational Systems



Buoyancy Engine

- Controls density of glider relative to seawater
- Bleeds down at surface, pumps at apogee/deepest part of dive and on way to surface
- Pumping is loud!



Attitude & Flight

- Internal mass shifter (battery) or external rudder controls pitch, roll, heading to keep glider on course
- Glider estimates its location underwater using dead-reckoning



Communications

- Two-way communication at the surface: **reports** engineering data and (some) science data and **receives** new commands
- After every dive, or every few dives



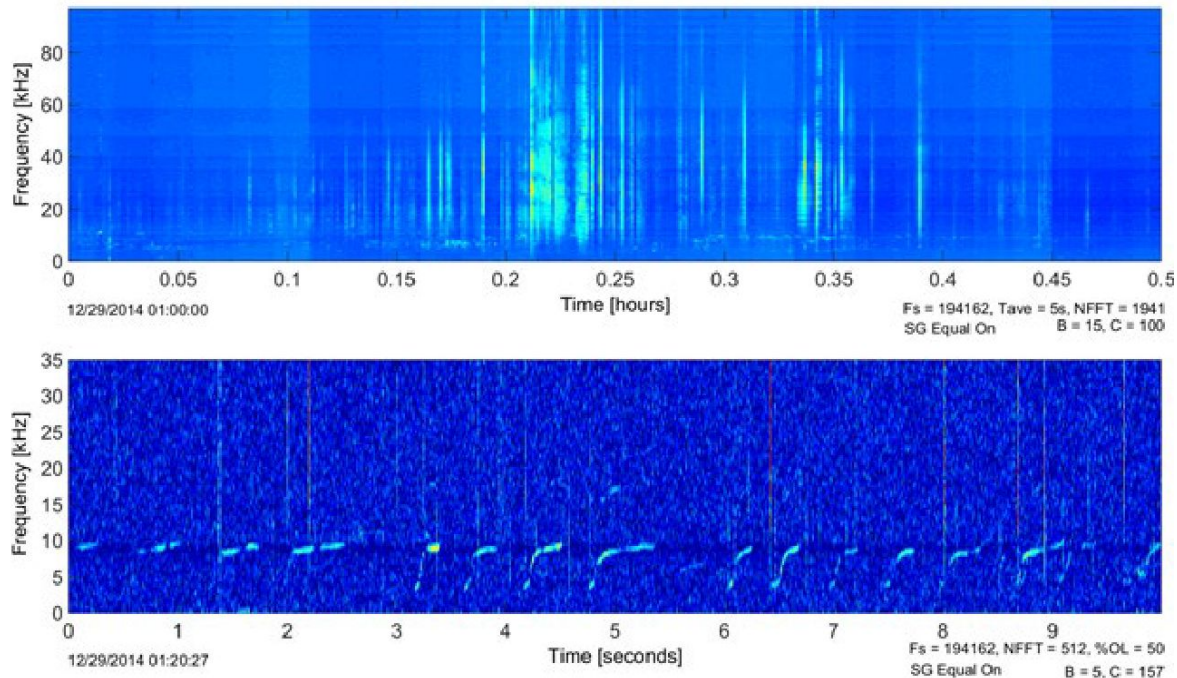
Science Payloads

- Range of sensor options
- CTD, fluorometry, pH, oxygen, turbidity, shadowgraph, AZFP, microstructure, **PAM**
- Sampling regime can be set by pilot

What does a pilot control?

- Flight path (waypoints, heading, current correction)
- Max dive depth
- Vertical speed (buoyancy)
- Pitch angle/glide angle (dive steepness)
- How often the glider should adjust internal motors (guidance and control)
 - Some gliders have “autoballast” that will adjust this automatically
- Number of yos/dives between surface fixes
 - And subsurface depth
- Drift depth and time
- When certain sensors operate
- Use of an altimeter

Why PAM?

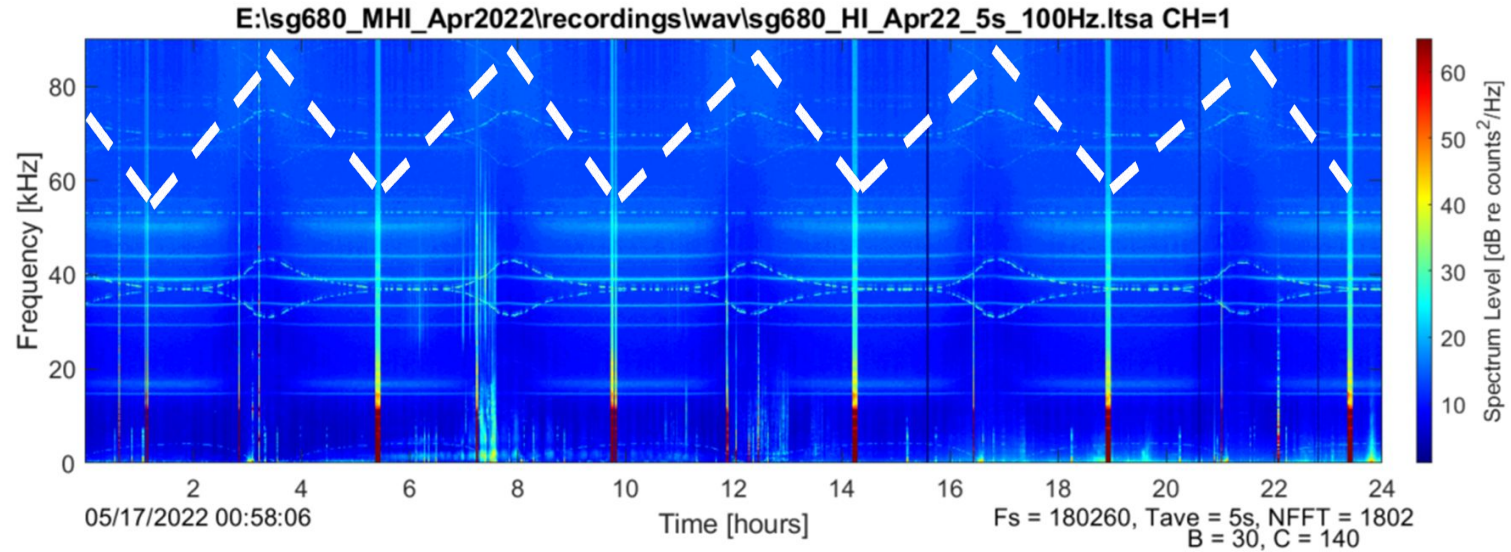


Gliders are generally very quiet, making them great for PAM data collection

Intermittent pumping and other motor controls do introduce self-noise

Low frequency flow noise can occur at certain speeds

Why PAM?



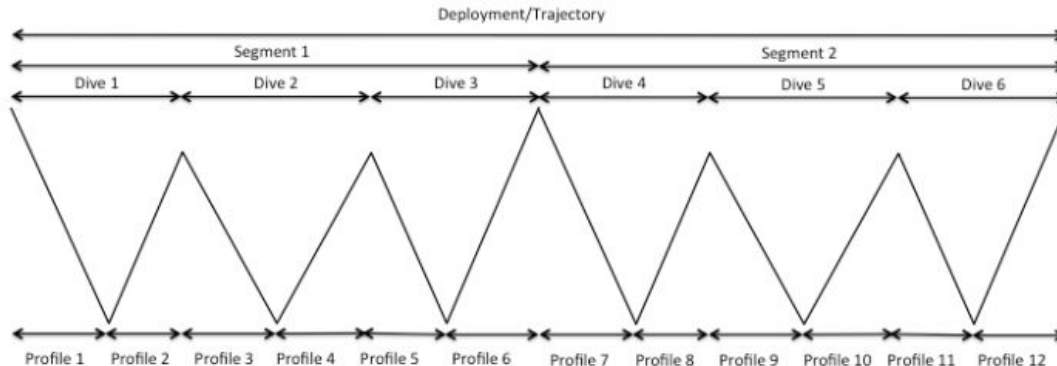
Gliders are generally very quiet, making them great for PAM data collection

Intermittent pumping and other motor controls do introduce self-noise

Low frequency flow noise can occur at certain speeds

Some terminology (from the IOOS NGDAC)

- **Profile:** single vertically oriented track, either downward (descent, downcast) or upward (ascent, upcast)
- **Dive:** A downward profile to max target depth followed by an upward profile to the surface or a shallow turnaround depth. Made of two sequential profiles (down + up)
- **Segment:** All data collected between two successive GPS surface fixes. This is at least one dive, can be multiple complete dives.
- **Trajectory:** Complete sequence of segments from initial deployment to recovery



Some more terminology

- **Loiter** or **drift** or **quiet mode**: When the glider turns off all motors and remains submerged at a specified depth for a specified period of time
-

Rodeo operations plan

Phase 1: shakedown/trimming. Get your glider working.

Phase 2: fast mode. Fly gliders with goal of making good progress over ground and against currents and staying near the trackline.

Phase 3: slow mode. Fly gliders with the goal of quiet acoustic data - reduced speed and motor adjustments.

Phase 4: intermediate mode. Balance progress along the trackline/maintaining course with minimal glider self-noise or flow noise. How you might fly a “typical” PAM mission.

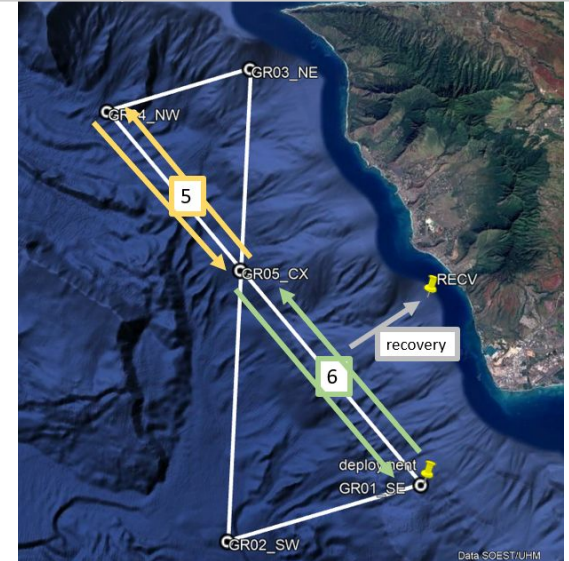
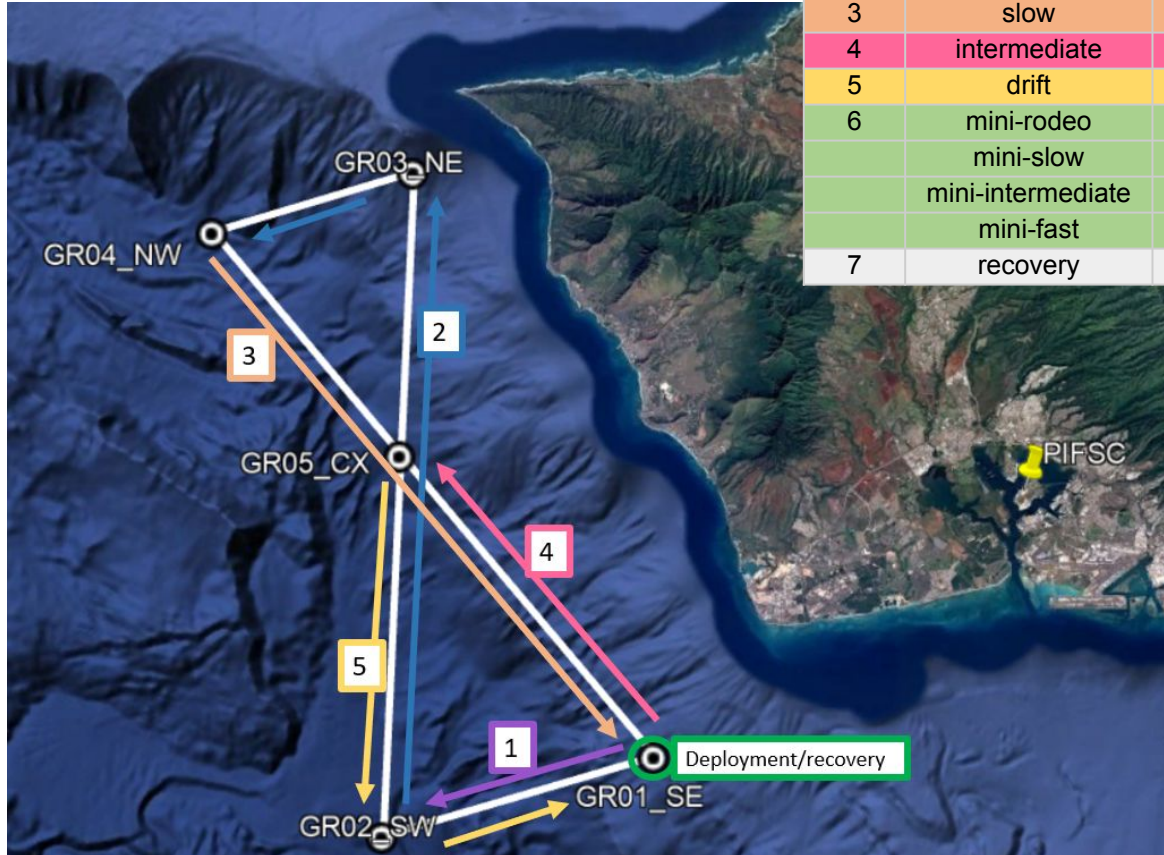
Phase 5: drift mode. Same settings as intermediate mode, but drift at 800 m depth for 20 min.

Phase 6: mini-rodeo. Cycle back through each above mode for ~12 hours (to account for SeaExplorer data logging issue). SLOW, INTERMEDIATE, FAST (DRIFT mode not repeated; last 12 hours used)

Phase 7: recovery

Rodeo operations plan

Phase	Mode	Distance (km)	Start Date (UTC)	Stop Date (UTC)
1	shakedown	18	2026-01-28 22:00	2026-01-30 06:00
2	fast	52	2026-01-30 06:00	2026-02-01 18:00
	shallow dives			
3	slow	42	2026-02-01 18:00	2026-02-04 18:00
4	intermediate	25	2026-02-04 18:00	2026-02-06 18:00
5	drift	43	2026-02-06 18:00	2026-02-08 18:00
6	mini-rodeo	variable	2026-02-08 18:00	2026-02-10 06:00
	mini-slow		2026-02-08 18:00	2026-02-09 06:00
	mini-intermediate		2026-02-09 06:00	2026-02-09 18:00
	mini-fast		2026-02-09 18:00	2026-02-10 06:00
7	recovery		2026-02-10 06:00	



Data overview

Folder for each glider contains semi-standardized timeseries outputs

- **GPS:** true locations from surfacings. May be every dive or every few dives
- **Science:** CTD data. Locations are interpolated or dead-reckoned underwater locations
- **Flight:** glider operational information. At a resolution similar to or the same as the CTD data (underwater, dead-reckoned positions)
 - Seaglider has engineering and modeled flight data - engineering data is measured, modeled data is output from the hydrodynamic model. All flight data for Slocums are combined into one timeseries
- **Detections:** at hourly and event-level granularity. Simulated dataset

Hardware Key:

<https://nmfs-pam-glider.github.io/GliderRodeo/content/hardware.html>

Acoustic Hardware

A total of 8 gliders from four different companies, with four different PAM sensors, were included in the PAM-Glider Rodeo. A summary of the platform specifications are provided in the table, below, and additional details are provided for the Glider platforms, acoustic sensors, and oceo sensors (below).

Show entries Search:

	Glider.Name	Owner	Type	Manufacturer	PAM.system	Glider.ID	Hydrophone.Type	Hydrop
	All	All	All	All	All	All	All	All
1	risso	SWFSC	Slocum G3S	TWR	DMON2	1025		65
2	stenella	SWFSC	Slocum G3S	TWR	WISPR 3.0	1031	HTI-99HF	85619€
3	Will	OSU	Seaglider M1	Kongsberg	WISPR 3.0	SG607	HTI-92WB	65300 dB)
4		PIFSC	Seaglider SGX	UW/IOP	WISPR 3.0	SG274	HTI-92WB	63505 dB)
5	belladonna	Cornell	Oceanscout	Hefring	WISPR 6.1	belladonna	HTI-92WB	
6	rosie	Cornell	Oceanscout	Hefring	WISPR 6.1	rosie	HTI-92WB	
7	SEA117	Alseamar	SeaeXplorer	Alseamar	Auris	SEA117	Auris	
8	capex987	Teledyne	Slocum G3S	TWR	JASCO Observer	capex987	M36-V35-EXBH-100, M36-V35-100, M36-V35-100	H0010, H0013, H0012, H0012

List of Gliders

Showing 1 to 8 of 8 entries Previous Next